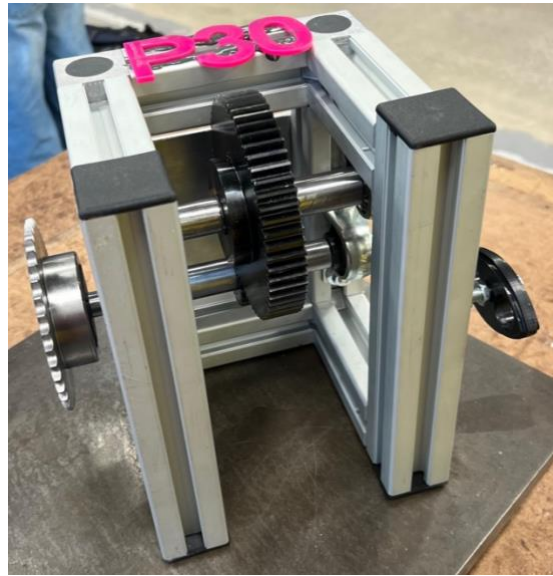


IMPERIAL

F24 Car Drive Transmission Project



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Executive Summary

The objective of this project was to design and manufacture a drivetrain connecting an electric motor to the rear axle of an F24 kit car (pictured below as Figure 1). The final product was a fully operational three-stage transmission enclosed within the provided assembly casing. The drivetrain successfully stepped down the speed of the motor with modification $G = 9.5$ to a linear velocity of $v = 20.1 \text{ km/h}$ well below the $v_{max} = 25 \text{ km/h}$ maximum constraint.



Figure 1. F24 Car (back view) with transmission inside

Steel EN1A, with additional cosmetics laser cut from acrylic. With the purpose of reducing the workload of the manufacturing phase, both shafts were manufactured using Computer Numerical Control (CNC) as duplicates and were then adjusted by turning on a lathe; taper bushes were also selected as axial constraints to remove the need to manufacture endcaps and keyways for the sprockets. Other manufacturing methods used included milling, drilling and fabrication for the assembly casing and gears.

The project was to be completed in the 10-week timeframe, of which the initial four weeks were dedicated to design and development. During this phase, the team composed design concepts and developed a detailed CAD model with Engineering drawings using SolidWorks. The main problem we faced during this time was interference between the proposed drivetrain and the vehicle chassis. This was solved by mounting the motor above the drivetrain casing, significantly improving packing.

Our team followed the provided design brief [1], ensuring all components fit within the dedicated space inside the chassis as to health and safety standards. All parts were either sourced from RS Components or manufactured in the Student Teacher Workshop using Aluminium 6082 T6 or Mild

Full experimental testing of the drivetrain–motor assembly could not be fully completed due to the cancellation of allocated testing slots and several STW sessions because of industrial action. Consequently, the final assembly was only completed shortly before the submission deadline, preventing some validation of performance, noise, or durability at this stage.

In hindsight, improvements could have been made by allowing greater contingency in workshop planning. Further optimisation of gear ratios and the inclusion of noise-reduction measures could also be explored in future work.

Overall, this project successfully delivered an efficient and functional drivetrain assembly that satisfied all performance, packaging and safety requirements stipulated in the design brief.

1. Introduction

This 10-week long project consists of designing and manufacturing a drivetrain for an F24 kit car. The objective was to step down the input speed of the electric motor to a maximum of $v_{max} = 25 \text{ km/h}$ on the rear wheel axle.

The final design is a 3-stage drivetrain of overall modification ratio $G = 9.5$. The first stage is a chain drive connecting the motor sprocket to the first shaft, then a gear drive connects the two shafts, and finally the second shaft is connected to the rear axle sprocket by another chain drive.

This report first covers the design process which includes the planning, calculations, ideas, concepts, product design specification and final design. Then, the functional and operational description of the drivetrain is made including design justifications and potential failures.

The respective roles of our 5 team members are listed below in Table 1.

Table 1. Team member roles

Member	Organisational Role	Engineering Role
Azaan Yaqub	Project Manager	Communicative and design lead
Antoine Buchet	Quality control officer	Transmission calculations specialist
Lakisanan Ratnarajah	Secretary	CAD drawing specialist
Hengze Li	Chair	CAD assembly specialist
Noor Dajani	Documentation officer	Manufacturing lead

2.1 Research and Product Design Specification

The brief gave a vague design description of a drivetrain for an F24, electric motor-powered car. Not much initial research was conducted to fill the PDS (Table 2 below) as aspects such as maximum speed, approved materials, manufacturing processes required, CAD drawing projections, and budget were strictly stated. Additionally, environmental responsibilities like minimizing material waste were not stated in the brief but by our design tutor during our first tutorial.

For more considerations, our group brainstormed and thought of more performance requirements. A major target for all automobile companies during design and manufacturing is to make their cars as energy efficient as possible to increase performance while minimising waste energy. We choose our target efficiency of 60% relatively lower than a quoted average efficiency for EVs of '77%' [2], as our technical expertise and manufacturing capabilities are lacking relative to larger producers. Noise was also a consideration, where we decided on a maximum level of '100dB' [3] according to research conducted (where 100dB is the max noise level ears can withstand for up to 15 minutes with no hearing damage).

Table 2. Product Design Specification

Category	Description	Validation
Performance Requirements:		
Speed	Maximum car speed of 25 km/h via adjusting transmission ratio. Target of 20km/h	Prototype testing measurement
Efficiency	Greater than 60%	Motor analysis
Noise	Maximum 100dB	Prototype testing measurement
Functional Requirements:		
Transmission	Transmit motion from motor to wheel axle with correct ratio and direction	Successful prototype testing
Ease of Implementation	Must be detachable and re-attachable from chassis in reasonable time frame (no welding)	Practically test (under timed conditions)
Compatibility	Compatible with rear frame and mounting points of F24 car	Successful prototype testing
Materials and Manufacturing:		
Approved Materials	Aluminium 6082 T6, Mild Steel EN1A, Polyoxymethylene, Acrylic	Review of bill of materials
Manufacturing Processes	Must use turning, milling and laser cutting a minimum of once. Maximum of 1 CNC part (duplicates are allowed)	STW workshop
Safety and Compliance:		
Covering	All rotating components fully enclosed; drivetrain should be fully protected	Safety viewing before prototype testing
Stability	Stable mounting, no loose or vibrating parts	Prototype testing observation
Supplier Limitations:		
Supplier	Only components found on RS Components	Review of RS form order
Budget	Maximum of 300 GBP	Review of RS form order
Environmental Responsibilities:		
Material Waste	Reduce quantity of wasted materials – plan material usage out beforehand	Observation of wasted parts

2.2 Decision Process

Firstly, we had to decide whether to use a chain drive or gearbox for our transmission using a pros and cons table (Table 3) and a subsequent scoring system (Table 4), where scores are on a scale from 0-10 and weighted.

Table 3. Chain drive vs Gearbox pros and cons

	Pros	Cons
Chain Drive	- Cost effective - High efficiency - Easy to replace - Direct power transfer	- Requires frequent maintenance - Wears and stretches over time - Requires tensioning
Gearbox	- More compact - Improved torque transmission - Lower motor torque requirement	- Very heavy - More costly - Can be excessive for DC motor

Table 4. Chain drive vs Gearbox scoring

Design	Replacement ease		Cost		Compactness		Total
	Score	Weight	Score	Weight	Score	Weight	
Chain Drive	8	0.50	6	0.25	4	0.25	6.5
Gearbox	3		4		8		4.5

For our scoring factors, we did not consider ease of maintenance as an F24 car will have very low usage rates (compared to daily drivers).

After deciding on a chain drive, our gear ratio was calculated by finding the angular velocity of the rear wheels, using a target velocity $v = 20 \text{ km/h} = 5.56 \text{ m/s}$, and wheel radius $r = 0.25 \text{ m}$ using equation 1 below. The selected DC motor has a nominal speed of 2100 rpm at operating voltage, and this was then used to calculate the gear ratio G using equation 2.

$$\omega_{\text{rear wheel}} = \frac{v}{r} * \frac{60}{2\pi} = 212 \text{ rpm} \quad (1)$$

$$G = \frac{\omega_{\text{motor}}}{\omega_{\text{rear wheel}}} = \frac{2100}{212} \approx 9.9 \quad (2)$$

A modification of 9.9 is quite substantial and would lead to unrealistically large sprockets that wouldn't fit inside measured casing dimensions (Figure 2). Therefore, an intermediate stage is needed to connect the motor sprocket and sprocket 1 to sprocket 2 and wheel axle sprocket (where G_n per stage = $\sqrt[3]{9.90} \approx 2.15$).

Due to limited spacing, a **gear stage** is more suitable as the intermediate stage instead of another sprocket stage (as shown in Figure 3) as the constrained space inside the metal casing prevents effective use of chains and large radii sprockets.

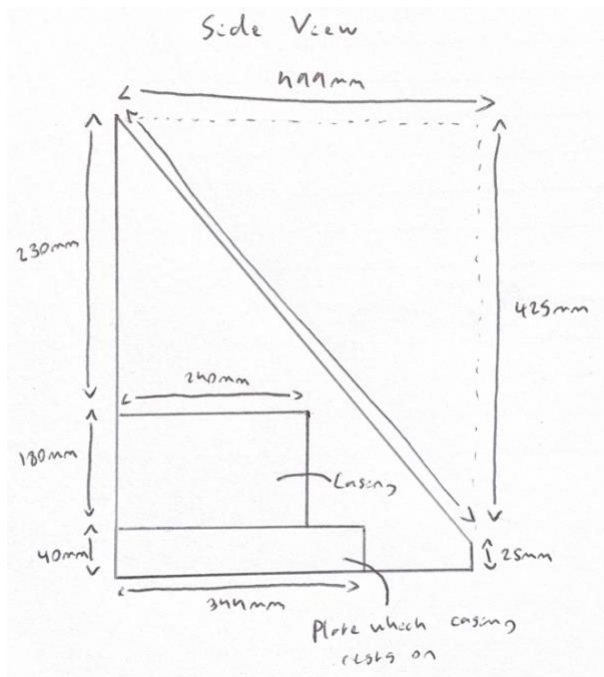


Figure 2: Measured dimensions of closed lid and casing

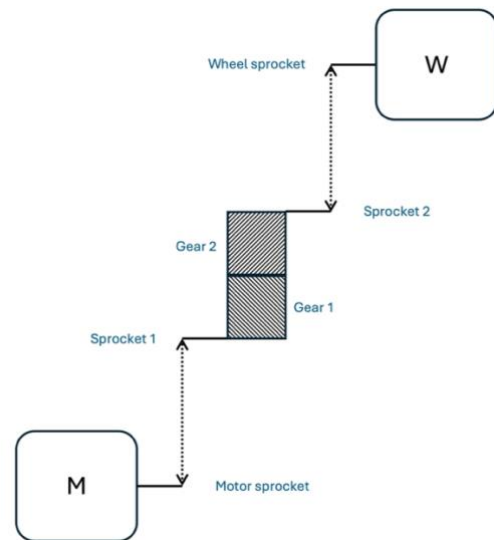


Figure 3: Updated transmission diagram

Hence a 3-stage transmission:

- **Stage 1:** Motor sprocket (fixed at 12 teeth) -> Sprocket 1
- **Stage 2:** Gear 1 -> Gear 2
- **Stage 3:** Sprocket 2 -> Wheel sprocket (fixed at 17 teeth)

We encountered an additional problem when viewing the CAD file for the car, where a steel beam impedes a direct chain path from the motor sprocket in its current position to the gearbox and casing where sprocket 1 would be located (Figure 4).

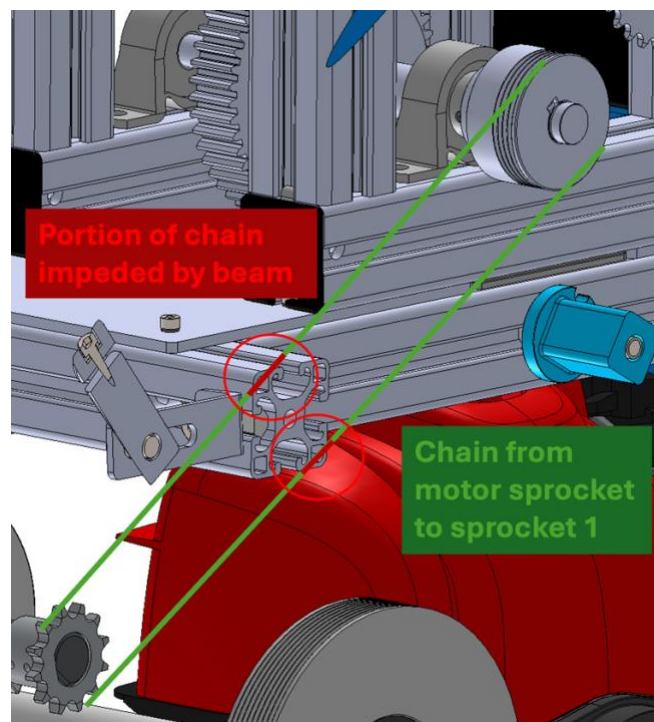


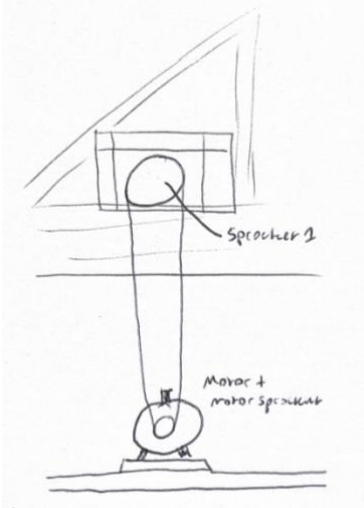
Figure 4. Annotated CAD assembly of impeding beam and drawn on chain

2.3 Design solutions

After brainstorming as a group, we came up with 4 proposed solutions in Table 5 to the problem of the impeding beam and how to connect the motor sprocket and sprocket 1 together.

Table 5. Proposed solutions

Design Description	Concept Sketch	Pros	Cons
Use 2 tensioners to funnel chain below the beam		- No need to move motor - Tensioners prevent derailment	- Takes up a lot of budget buying 2 tensioners - Difficult to assemble and position both tensioners correctly
Loop chain around beam		- No need to move motor - No additional parts needed to be bought or manufactured	- Bad design practice to have beam inside the path of the chain - Would need a large sprocket 1 to ensure chain is able to loop around beam without contact
Move motor on top of metal frame and funnel chain right below to sprocket		- On the frame so would be easy to detach and reattach whole design - Short chain length so less material waste and easier to tension via removing links	- Requires a motor plate and adapter to be made - No tensioner so still could result in chain derailment

Move motor to bottom of car and funnel chain upwards		- Prebuilt adapter at bottom of car already so no need to make a motor plate - No parts within chain so good design practise	- No tensioner and very long chain so could very easily derail - Difficult to attach and reattach as motor
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We subsequently selected the final design using a design matrix, with factors corresponding to key points in our PDS, and a scoring system (Table 6), like Table 4.

Table 6. Decision matrix for proposed solution

Design	Manufacturing ease		Installation ease		Safety		Reliability of tensioning		Total
	Score	Weight	Score	Weight	Score	Weight	Score	Weight	
2 tensioners	2	0.20	3	0.20	5	0.35	9	0.25	5.00
Chain around beam	8		4		1		3		3.50
Motor on top of frame	5		8		8		7		7.15
Motor at bottom of car	7		6		6		6		6.20

We decided to go forward with 'Motor on top of frame' design as it is relatively easy to manufacture and install. Also, it does not have any major issues with chain tensioning so it should be reliable when testing in the summer term (as less chance of chain derailment).

3.1 Finalising Design Details

Once the outline for our final design was selected, we decided on the types of parts (gears and sprockets) we would select. Eventually, we picked:

- **Gears -> Spur Gears:** have a higher power transmission efficiency ranging from '95% to 99%' [4] compared to helical gears
- **Sprockets -> Taper bushing Sprockets:** save manufacturing time (compared to using a sprocket with no taper bush) as the taper bushing axially restrains the sprocket. Hence, we do not need to use keys or endcaps so saves a lot of manufacturing time (by avoiding the need to mill keyways and threaded holes)

Additionally, we chose to make our 2 shafts near identical in length and diameter at each step. This means that, as we can only CNC duplicate parts, we can CNC both shafts – saving lots

of manufacturing time by not needing to turn down an entire shaft manually. But shafts still need to be different as sprockets on ends have different distances at which they need to be aligned axially. So, the solution is to keep the step lengths and diameters identical bar the last two steps (which are outside of the 2 bearings and therefore not within the frame).

As can be seen in Figure 5 below, we have purposefully made shaft 1 longer so that it is possible to cut and turn this shaft into shaft 2, acting as a good design compromise as minimal manufacturing time is needed to alter one shaft to the degree we require.

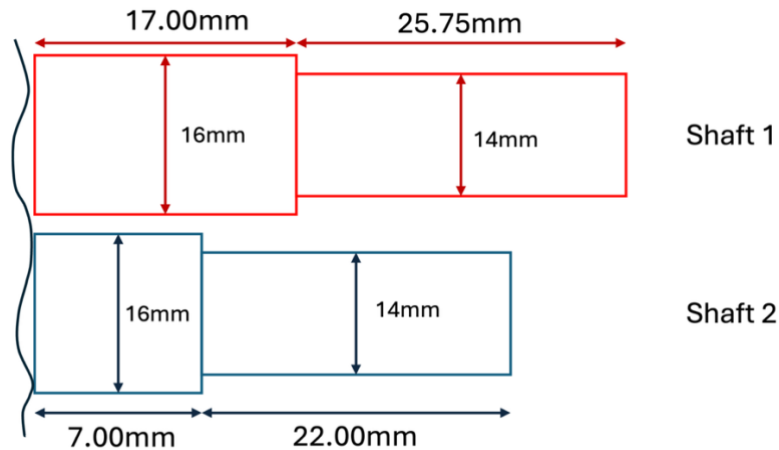


Figure 5. Diagrammatic representation of differences of ends of shafts 1 and 2

Finally, as the distance (and hence respective chain lengths) for stage 3 is much greater than that of stage 1, a tensioner is needed in between sprocket 2 and the wheel sprocket. Because we had very minimal spare budget at this point in the part selection process, we could either manufacture our own tensioner (with the requirement of purchasing an extra bearing and sprocket) or purchase a chain tensioner (idler tensioners were out of our budget).

Due to our focus of reducing the manufacturing time of our transmission, we chose to purchase a chain tensioner from RS orders. This tensioner will be attached to the side groove of our frame (as we are not allowed to alter the rest of the F24 car) using an inverted T-bolt and nut (Figure 6).

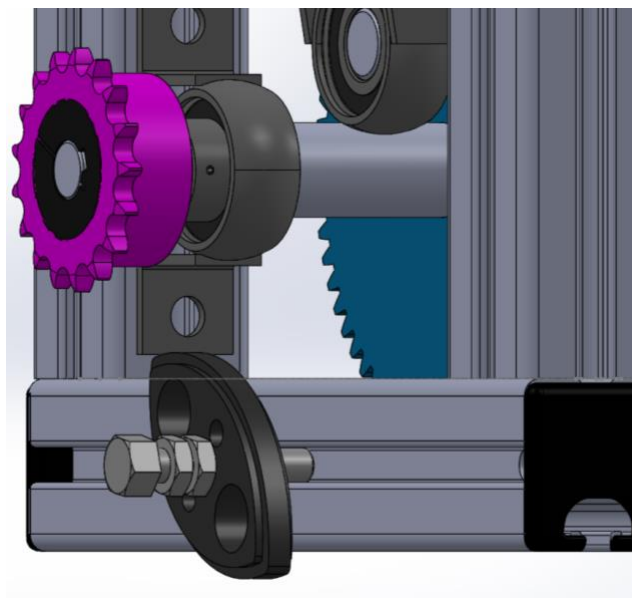


Figure 6. CAD image of tensioner attached to transmission frame

3.2 Engineering Analysis

Actual Gear Ratio Calculations:

As taper bushing sprockets need matching taper bushings, they have less variety in teeth number on RS orders. Therefore, our resulting gear ratio is not exactly 9.9 (as shown in Equation 3).

Stage 1: 12-tooth motor sprocket driving a 38-tooth sprocket.

Stage 2: 20-tooth gear driving a 60-tooth gear.

Stage 3: 17-tooth sprocket driving a 17-tooth sprocket on the rear axis.

$$G_{actual} = \frac{38}{12} * \frac{60}{20} * \frac{17}{17} = 9.5 \quad (3)$$

Whilst 9.5 leads to a higher linear velocity than the target 20km/h, we did not account for friction between the tires and surface the car runs on. Thus, the car will be slower than calculated, and closer to target 20 km/h value.

A dynamic acceleration model (developed in Python) was used to predict the vehicle speed–time response and verify that the drivetrain meets the 25 km/h speed limit. The motor was modelled using the linear torque–speed relationship (Equation 4 below):

$$T(n) = 11.5 - 0.005n \quad (4)$$

Here, T is the motor torque (Nm) and n is the motor speed (rpm). The motor torque was multiplied by the overall gear ratio and drivetrain efficiency (taken as 90%) to obtain the wheel torque, and the resulting tractive force at the tyres was calculated using the wheel radius (0.25m measured on car using ruler). Resistive forces were included in the model as rolling resistance and aerodynamic drag.

The net force acting on the vehicle was then used in Newton’s second law to compute acceleration, which was numerically integrated over time to obtain the vehicle velocity and produce the speed against time graph below (Figure 7).

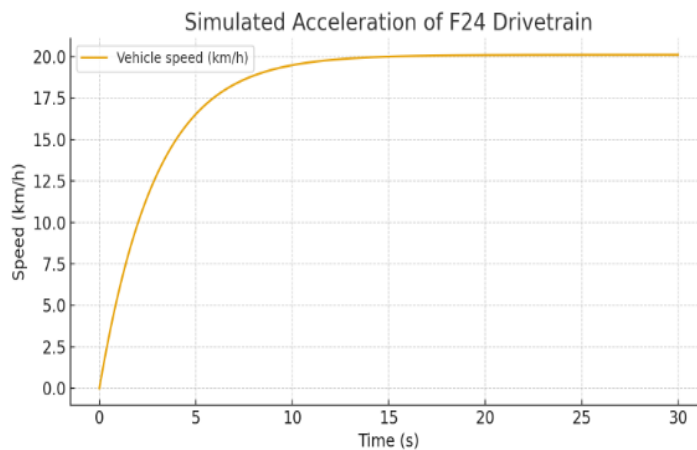


Figure 7. Speed and time graph modelling dynamic acceleration (on Python)

The graph in Figure 6 confirms that, including resistive forces, the car will reach our target speed of 20 km/h despite $G_{actual} < G_{calculated}$.

Shaft Calculations:

Both shafts experience their highest torque when the vehicle starts from rest (stall torque $T_{stall} = 11.5 \text{ Nm}$). According to the transmission ratios, the second shaft (the shaft carrying the 60-teeth gear and 15-teeth sprocket) is subjected to the largest torque (as shown in Equation 4).

$$T_{max} = T_{stall} * G_1 * G_2 = 109.25 \text{ N} \quad (4)$$

The minimum required shaft diameter is found from the torsion relationship. We can use the torque equation (Equation 5 and 6):

$$\frac{\tau}{r} = \frac{T}{J} \quad (5)$$

Where:

$$J = \frac{\pi}{32} (d^4) \quad (6)$$

This can be rearranged to give an expression for the shaft diameter in terms of the applied torque and allowable stress (Equation 7), where substituting the maximum torque on the second shaft and the design shear stress (330 MPa for steel [5]) gives:

$$\frac{r}{\frac{\pi}{32} ((2r)^4)} = \frac{\tau}{T} \quad (7)$$

$$r_{min} \approx 5.95 \text{ mm}, d_{min} \approx 11.90 \text{ mm}$$

This required diameter is less than the minimum shaft diameter selected (14 mm); therefore, the shafts are adequately strong in torsion.

Even though there is only a safety factor of 1.18, this is assuming that the transmission is fully efficient whereas in reality T would have a much smaller value (as $G_{actual} < G_{calculated}$) and therefore yield a lesser value for d_{min} .

Gear minimum face width Calculations:

As the 20T gear is the smallest gear, it is most likely to break first and so is treated as the critical component.

Both gears are made from EN8 steel, for which the permissible bending stress is $\sigma_p = 131 \text{ MPa}$ [6]. Lewis form factors taken from the data tables give $Y = 0.30769$ for the 20T gear [4]. The tangential force applied on the gear teeth could be obtained from the power transmitted through this stage using Equation 8 below.

$$F = \frac{P}{v} = \frac{P_{nominal}}{\omega r} = 172.8 \text{ N} \quad (8)$$

Where v is the linear speed of gear teeth, ω is gear rotational speed in rpm and r is half PCD.

The dynamic factor K_v is (Equation 9):

$$K_v = \frac{6.1}{6.1 + v} = 0.815 \quad (9)$$

Using the fact that our selected gear module is 2, the gear minimum face width (Equation 10) yields:

$$w = \frac{F \times 10^6}{K_v * m * Y * \sigma_p} = 2.63 \text{ mm} \quad (10)$$

In the final design, a face width of 20 mm is specified for this gear pair. This is more than seven times the minimum required value, so the gears have a large safety margin against bending failure. Also, this cross-sectional area means that the gears will continue to maintain near-full contact despite not being axially aligned perfectly.

Bearing Calculations:

Pillow-block ball bearings with a 17 mm bore are used to support the shafts.

According to the manufacturer's catalogue, the selected bearing has a dynamic load rating of $C = 13.6 \text{ kN}$. The bearing next to the 17T sprocket is expected to be the most heavily loaded as it is on the 2nd shaft which has the greatest torque. A basic rating life of 10 hours of operation is assumed. The corresponding life in revolutions is (Equation 11):

$$L_{10} = L_{10h} \times \omega \times 60 = 10000 \times 2100 \times \frac{12}{38} \times \frac{20}{60} \times 60 = 132.6 \times 10^6 \quad (11)$$

Where ω is the angular speed of the bearing.

The radial load on this bearing is taken to be equal to the greatest torque applied to the shaft. The axial load is neglected as it is much smaller than the radial load. Therefore, load factors $X = 1$ and $Y = 0$ are selected [4]. The maximum radial load on one bearing (2 bearings per shaft) is subsequently obtained below using Equation 12:

$$F = XF_r + YF_a = 1 \times F_r = \frac{T}{2 \times r} = \frac{T_{stall} \times \frac{38}{12} \times \frac{50}{20}}{2 \times r} = 1011.6 \text{ N} \quad (12)$$

Where T is the maximum torque applied to a bearing and r is its pitch radius.

So, the minimum dynamic load of bearing is (Equation 13):

$$C_{min} = F(L_{10})^{\frac{1}{k}} = 5.3 \text{ kN} < C_r = 13.6 \text{ kN} \quad (13)$$

The bearing is therefore suitable for this drivetrain application.

Operating point:

Figure 8 below illustrates the torque–speed characteristic of the 24 V green power motor. The relationship between torque T and motor speed n can be approximated by the same linear expression used in the dynamic approximation model above (Equation 4)

$$T(n) = 11.5 - 0.005n \quad (4)$$

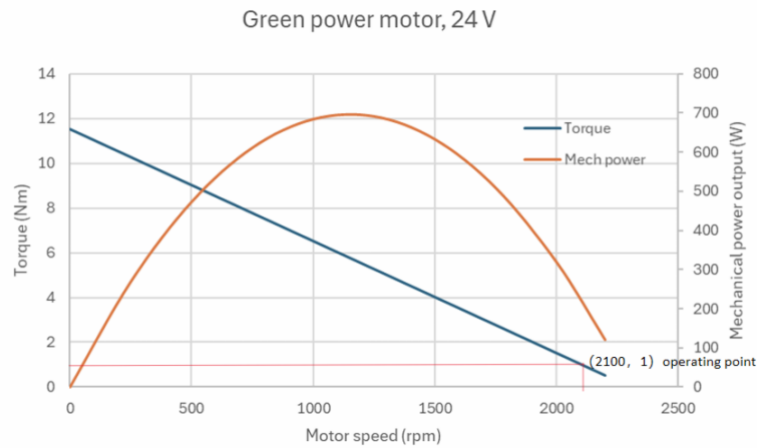


Figure 8: Torque against motor speed diagram. See [6]

We are targeting to run the motor at nominal motor speed of 2100 rpm, yielding operating point at (2100,1) on the diagram (traced by red lines). Substituting the nominal motor speed $n = 2100$ rpm into Equation 15 gives $T(n) \approx 1N$, hence linear expression corresponds to the diagram, and our motor operating point is confirmed.

3.3 Modes of Failure and Mitigation

A failure modes and effect analysis was conducted below (Table 7), allowing us to prioritise actions to eliminate major sources of failure.

Table 7. Failure modes and effect analysis for F24 transmission

Failure	Severity of failure	Likelihood of occurrence	Likelihood of detection	Risk Priority Number
	S	O	D	S*O*D
Incorrect motor direction	8	2	1	16
Insufficient lubrication	5	3	5	75
Gear misalignment	7	3	5	105
Chain derailment (primarily 1 st chain)	9	6	3	162

Based on the preceding analysis, chain derailment has been identified as the most critical and severe failure mode. A derailed chain not only interrupts power transmission but may also become entangled within the drivetrain, potentially causing the wheels to lock. Such an abrupt and uncontrolled deceleration poses a significant safety hazard to both the driver and nearby bystanders. So, this failure mode requires the most rigorous mitigation measures.

Initially we thought of two ways to mitigate this risk:

1. Stack penny washers underneath the motor, raising its height and hence increasing tension in chain as distance of sprockets increase
2. Remove links in chain used

However, detaching and repositioning the motor during testing, along with keeping spare penny washers, would be impractical. Therefore, removing chain links is the more feasible and effective method for maintaining appropriate chain tension.

4. Assembly Methods and Manufacturing

Manufacturing Methods:

Table 8 below describes the available manufacturing methods.

Table 8. Available manufacturing methods

Method	Description	Limitations	Potential Applications
Turning	Produces cylindrical parts by rotating the workpiece on a lathe with a stationary cutting tool	Time-intensive, can only be used for cylindrical parts	Shaft turning, gear bores, spacers, chamfers and fillets
Milling	Creates features using a rotating cutting tool on a stationary workpiece	Time intensive, only suitable for small parts	Shaft keyways
Fabrication	Involves cutting, bending and assembling components	Potentially less precise than other methods	Motor mounting plate, gear keyways, assembly
Computer Numerical Control (CNC)	Automated precision manufacturing	Limited to one part as per brief	Shaft turning, keyways, gear bores, motor mounting plate
Laser Cutting	Precise process to cut or etch into non-reflective materials	Not suitable for metals	Motor mounting plate, aesthetic additions

Manufacturing Processes:

Table 9 below shows the manufacturing process carried out, with issues we encountered and their respective solutions.

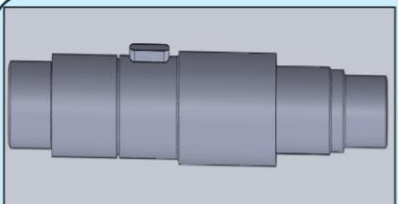
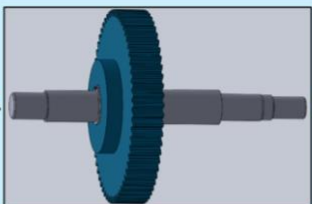
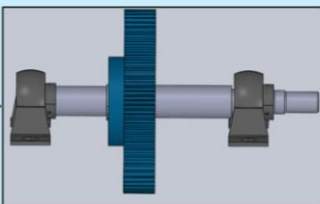
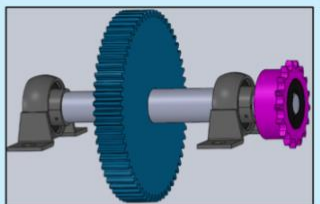
Table 9. Conducted manufacturing processes and their issues + solutions

Parts	Process	Issues	Solutions
Shaft body	- Both shafts CNC milled - Milled keyways with drill machine and slot drill attachment For only one shaft: - Cut the length of 14mm step by 13.75mm using lathe attachment - Turned the last 10mm of length of the 16mm step down to 14mm	- 13.75mm of steel is large amount of material to cut in one go - Difficult to find position where keyway needed to be drilled	- Faced off 1mm from shaft and repeated 13 times to achieve desired cut - Used edge finder (considering its 5mm radii) to zero and set lengths of shaft on drill machine
Spur gears	- Purchased on RS - Bored the internal diameters to correct dimensions - Broached 2.3mm keyways into gears	- Gear boss was too small to fix into lathe	- A smaller vice was installed to fit the boss diameter
Motor plate	- Cut 5mm aluminium sheet to correct dimensions - Marked drilled hole locations - Drilled clearance holes for motor mount and frame	- Holes need to be aligned accurately so that they match up with the motor mount and frame	- Clearance holes were used to give some allowance for machining error
Keys	- Purchased on RS components - No modifications	N/A	N/A
Sprockets & taper bushings	- Purchased on RS components - No modifications	N/A	N/A
Bearings	- Purchased on RS components - No modifications	N/A	N/A

Assembly Methods:

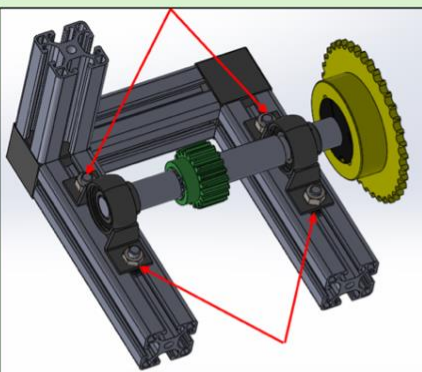
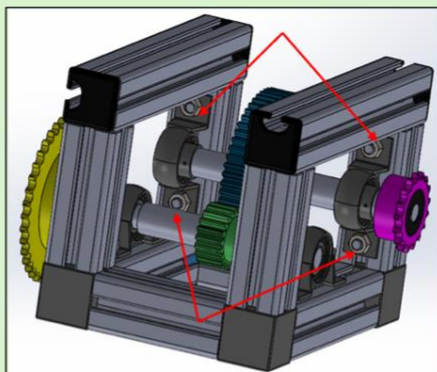
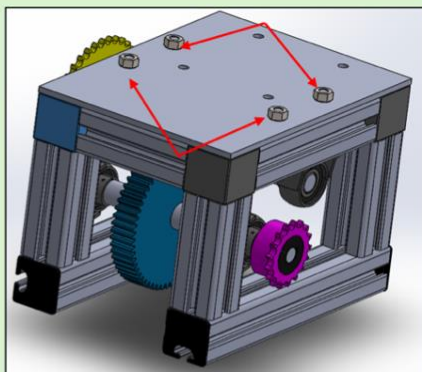
The two sections below visually illustrate the step-by-step assembly process.

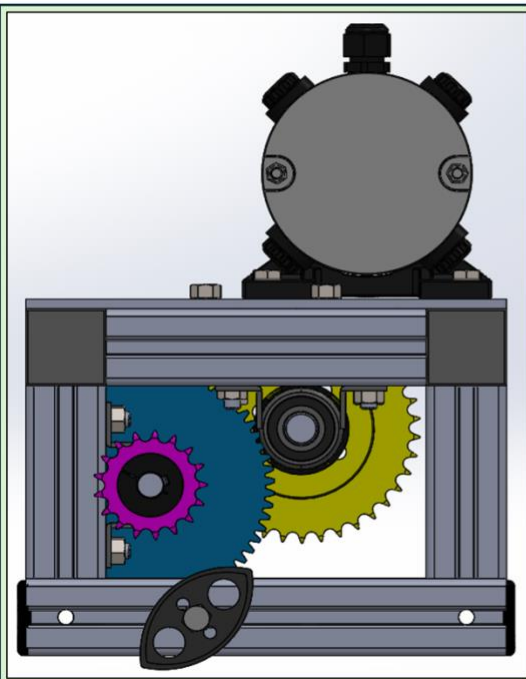
Subassembly of Shaft 2:

			
1. Prepare and Orient the Shaft: • Clean the shaft to remove any oil or burrs. • Orient the shaft so the circlip groove and keyway are clearly visible. • Insert the parallel key fully into the keyway. Ensure it sits flush and does not protrude above the shaft surface	2. Install the 60-Tooth Gear: • Align the keyway of the 60-tooth gear with the key inserted on the shaft. • Slide the gear onto the shaft until the gear boss sits just before the circlip groove. • Using circlip pliers (or tweezers if needed) fit the circlip into the groove to retain the gear. • Confirm the circlip is fully seated around its entire circumference.	3. Insert Bearings into Their Seats: • Identify the two bearing seats on the shaft. • Lightly lubricate the shaft to ease bearing installation. • Slide each pillow-block bearing onto the shaft and seat them against their respective shoulders. • Ensure each bearing rotates freely and is not loaded axially by the adjacent components.	4. Fit Taper Bush and 38T Sprocket: • Slide the taper bush onto the shaft. • Position the 38-tooth sprocket over the bush, aligning the bolt holes. • Insert and tighten the grub screws evenly to draw the bush into the sprocket and lock it onto the shaft. • Confirm the sprocket is fully seated and has no radial play.

Important to note that shaft 1 will have an identical assembly process than shaft 2 above.

General Assembly:

		
1. Install Shaft 1 and Bearings onto the Cassette Frame: • Position the cassette frame as shown in the assembly drawing. • Insert Shaft 1 (preassembled with gear/sprocket/bearings) into its designated bearing slots. • Secure the bearings to the cassette using T-head bolts and M8 nuts at the locations indicated by the red arrows. • Tighten the bolts evenly to ensure the bearings seat correctly and the shaft rotates freely.	2. Install Shaft 2 and Complete the Cassette Assembly: • Position Shaft 2 in the remaining bearing locations on the cassette. • Again, secure the bearings using T-head bolts and M8 nuts at the red-arrow mounting points. • Verify that the 20-tooth and 60-tooth gears are correctly meshed, with proper backlash and no binding. • Check for any interference between rotating components and the cassette frame before proceeding.	3. Attach the Motor Mounting Plate: • Align the motor plate with its mounting slots on the cassette. • Fasten it in place using T-head bolts and M8 nuts (locations shown by red arrows). • Ensure the plate sits flush and rigid, as this will determine the chain alignment during motor installation

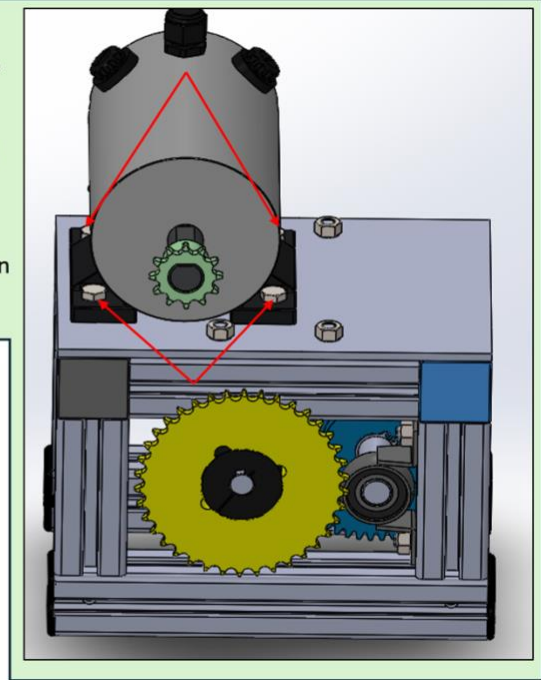


4. Mount the Motor onto the Frame:

- Position the motor on the mounting plate and align its fixing holes with the T-slot locations marked by the red arrows.
- Secure the motor using T-head bolts and M8 nuts, tightening them evenly so the motor sits flat and rigid.
- Verify that the motor sprocket is parallel and correctly aligned with the sprocket on Shaft 1 to ensure smooth chain engagement and prevent derailment.

5. Install the Chain Tensioner:

- Attach tensioner to cassette frame in orientation shown in the assembly diagram.
- Adjust its position so that the idler sprocket is parallel to both chain sprockets, preventing misalignment during operation.
- Set tensioner so sufficient chain tension to avoid derailment, while ensuring chain still moves smoothly.



5. Project Management

Below is a Gantt chart tracking progress (Table 10), and an individual time log (Table 11).

Table 10. Gantt Chart of Project Milestones

Task Description	Progress	Week of Project											
		1	2	3	4	5	Design Gateway	6	7	8	9	10	
Design													
Project Role Allocation	100%												
Project Design Specification	100%												
Concept Design	100%												
Transmission calculations / selection	100%												
Final Design	100%												
Component Order and CAD													
Part Modelling / Downloading	100%												
Assembly Model	100%												
Part / Assembly Drawing	100%												
RS Order Form	100%												
Bill of Materials	100%												
Manufacturing													
STW Session 1	100%												
STW Session 2	100%												
STW Session 3	0%*												
STW Session 4	100%												
STW Session 5	100%												

*Cancelled due to strikes

7. Conclusion

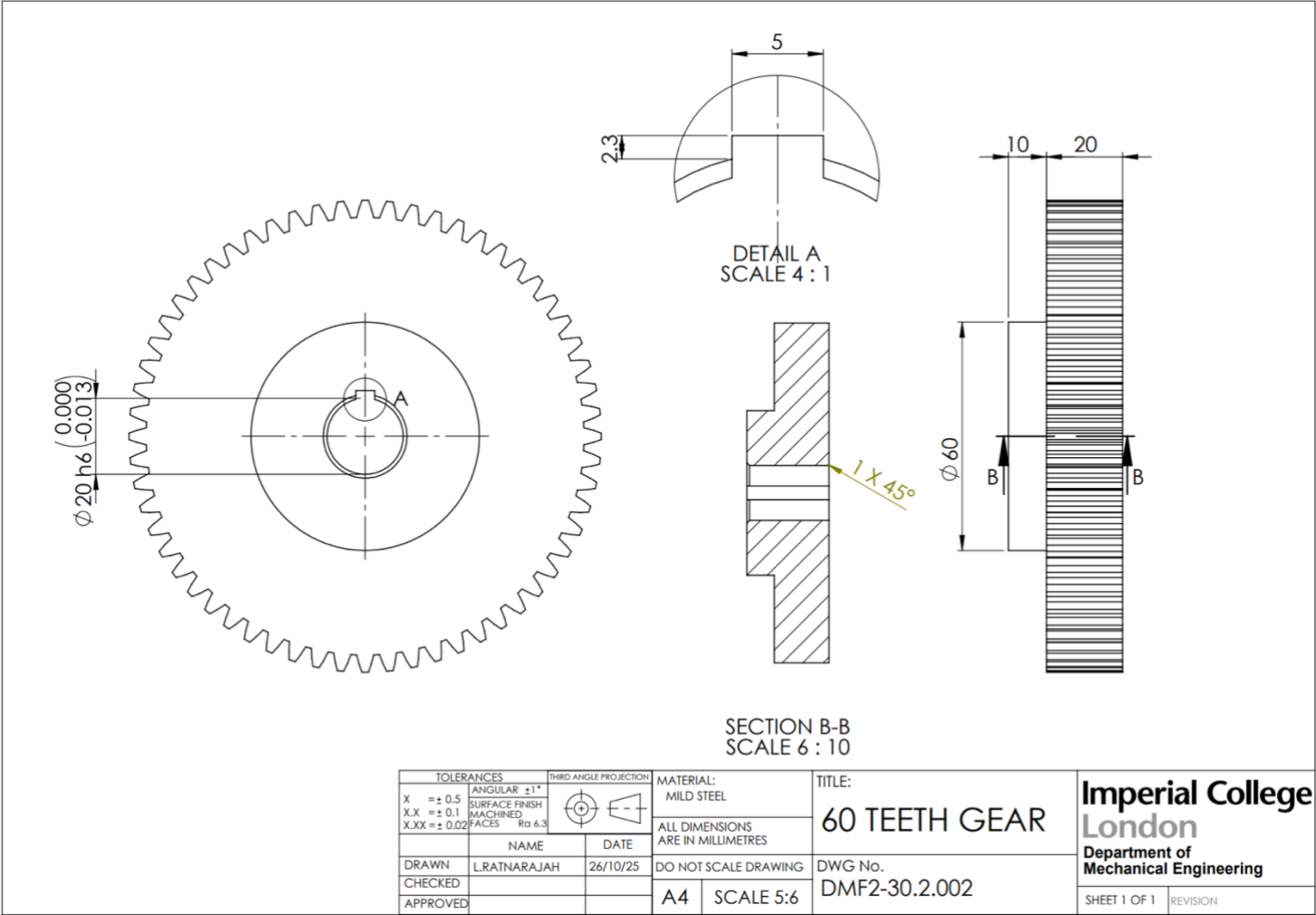
This project successfully achieved the design and manufacture of a drivetrain transmitting power from a 24 V DC motor to the rear axle of an F24 kit car. The design was developed in accordance with the project brief, which imposed a maximum vehicle speed of 25 km/h, a £300 component budget limited to RS Components, and the requirement for at least one part to be manually turned, milled, and laser cut. The project was completed by a five-member team through concept generation, engineering analysis, detailed design, and final manufacture of the drivetrain assembly.

The final solution is a three-stage transmission with an overall gear ratio of $G = 9.5$. The first stage consists of a chain drive with a 3.1:1 ratio connecting the motor to the first shaft. This is followed by a 3:1 spur gear stage transmitting torque to the second shaft, and a final 1:1 chain drive connecting the drivetrain to the rear axle. The entire system is mounted within an aluminium casing designed to allow the vehicle's rear lid to close fully. A chain tensioner was incorporated to prevent slack in the second chain. For manufacturing efficiency, one shaft was CNC-machined and duplicated, with the second shaft obtained by manual modification. All remaining components were sourced from RS Components, with additional manual turning and milling applied to gears and keyways. The drivetrain has now been fully assembled.

As physical testing will take place in the summer term, the current evaluation is based solely on analytical modelling. Future work should therefore focus on experimental validation of the vehicle's top speed, assessment of drivetrain stability under load, and measurement of operational noise to confirm compliance with the original design requirements.

8. Appendix A: Engineering Drawings

A.1: 60T Gear Part Drawing:



The drawing shows Shaft 2 with the following features and dimensions:

- Front View:** Total length 105 mm. Keyway on the left with a width of 15 mm (+0.1/0.0) and a depth of 7.5 mm. A central section with a diameter of 22 mm. A keyway on the right with a width of 16 mm (+0.005/-0.007) and a depth of 5 mm (-0.03/0.00). Surface finish symbols A and B are indicated.
- Top View:** Shows the circular profile of the shaft with a diameter of 17 mm (+0.0/-0.1). A keyway is shown with a width of 1.3 mm (+0.14/0.00) and a depth of 0.00 mm (-0.13/0.00). Surface finish symbol A is indicated.
- Detail A:** A circular detail view of the keyway on the left, showing a diameter of 18.61 mm (+0.00/-0.13) and a width of 1.3 mm (+0.14/0.00). Surface finish symbol A is indicated.
- Section B-B:** A cross-sectional view of the shaft showing a diameter of 16 mm (+0.005/-0.007) and a keyway with a width of 5 mm (-0.03/0.00) and a depth of 0.00 mm (-0.13/0.00). Surface finish symbol B is indicated.

Table 1: Tolerances

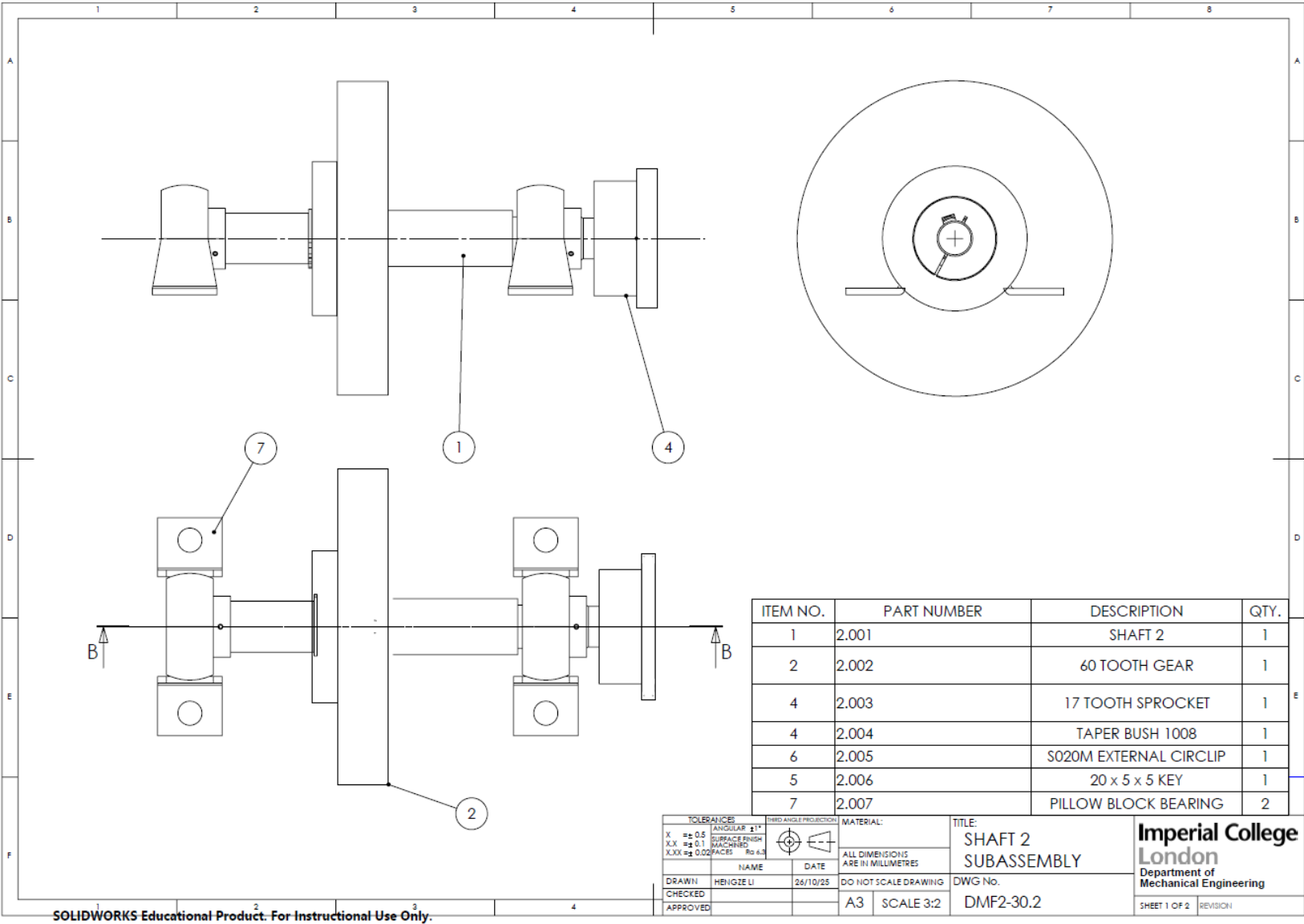
TOLERANCES	THIRD ANGLE PROJECTION
X = ± 0.5	ANGULAR ± 1°
X.X = ± 0.1	SURFACE FINISH
X.XX = ± 0.02	MACHINED FACES Ra 6.3

Table 2: Drawing Information

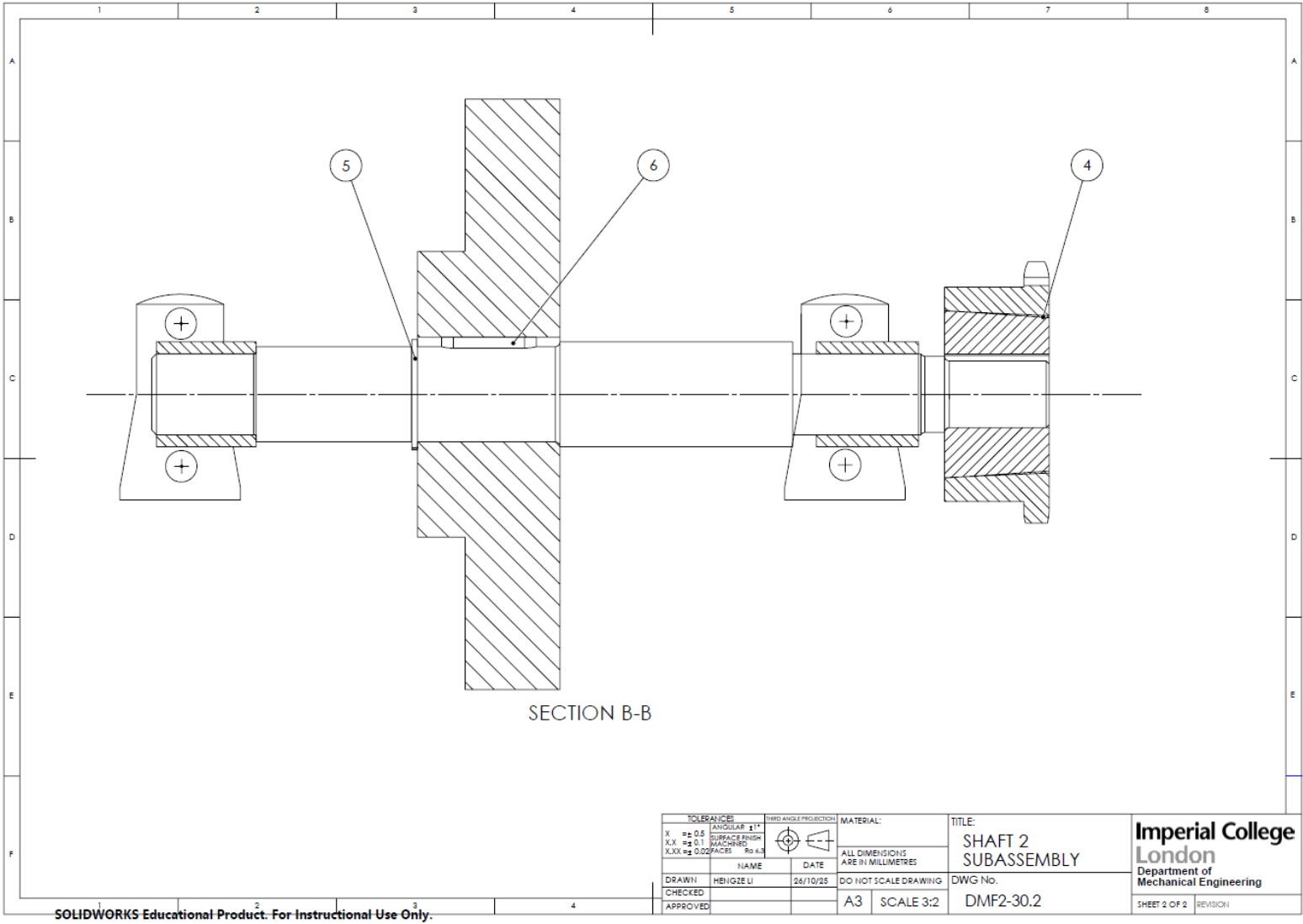
DRAWN		CHECKED		APPROVED		MATERIAL:		TITLE:	
L.RATNARAJAH	26/10/25					MILD STEEL EN1A	SHAFT 2		
DO NOT SCALE DRAWING						DWG No. DMF2-30.2.001			
A4						SCALE 5:4			

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A.3: Shaft 2 Subassembly Drawing (Pg.1):



A.3: Shaft 2 Subassembly Drawing (Pg.2)



Appendix B: Bill of Materials (BoM – Table 12):

Table 12. Bill of Materials

Item No.	Part No. (DMF2-30.)	Part Description	Material	Quantity
1	1.001	Shaft 1	Mild Steel EN1A	1
2	1.002	20 teeth Module 2 Spur Gear	Steel EN8	1
3	1.003	38 teeth Sprocket	Steel	1
4	1.004	Taper Bush 1210 Ø 14 mm	Cast Iron EN 1561	1
5	1.005	S020M External Circlip	N/A	1
6	1.006	20 x 5 x 5 Key	Steel	1
7	1.007	Pillow Block Bearing Ø 17 mm	N/A	2
8	2.001	Shaft 2	Mild Steel EN1A	1
9	2.002	60 teeth Module 2 Spur Gear	Steel EN8	1
10	2.003	17 teeth Sprocket	Steel	1
11	2.004	Taper Bush 1008 Ø 14 mm	Cast Iron EN 1561	1
12	2.005	S020M External Circlip	Steel	1
13	2.006	20 x 5 x 5 Key	Steel	1
14	2.007	Pillow Block Bearing Ø 17 mm	N/A	2
15	3	Gearbox Frame Assembly	Aluminium	1
16	4.001	Motor	N/A	1
17	4.002	Motor plate	Aluminium 6082 T6	1
18	4.003	12 teeth Sprocket	Steel	1
19	5	Tensioner	Galvanized steel	1

Appendix C: Risk Assessment (Table 13):

Table 13. Risk Assessment (for manufacturing and prototyping risks)

Hazards	Who it can harm	How it can harm	Control measures
Manufacturing Risks			
Contact with sharp edges after cutting	Anyone who touches the part	Sharp edges can cut hands	Chamfer rough edges via machine or manual processes
Drilling injury from slipping workpiece	Drill user and bystanders	Part could unfix and fly off due to speed of drill	Use clamps and tighten via hammering
Lathe entanglement	Lathe user and bystanders	Items can get caught in the lathe and fling off into surroundings	Avoid gloves, secure long hair, remove jewellery
Loose debris harming eyes	User and bystanders	Debris can fly into people's eyes	Always wear safety glasses
Finger injuries from saw	Saw user	Can press down saw when finger / hand in the way and cut it	Keep hands away from saw when in use, never touch the blade even if off
Prototype Testing Risks			
Incorrect motor direction	Bystanders	Car could start in reverse and run into people behind	Trial motor direction before attaching chain, reverse if necessary
Thermal expansion	Driver	Expansion could cause misalignment of parts and halt transmission (abruptly stopping car)	Ensure floating bearing has enough free space to account for expansion

Insufficient lubrication	Driver	Accelerates wear and can damage transmission and lead to lots of vibrations (shaky drive)	Lubricate chains and gears before testing
Gear misalignment	Driver	Reduced power transmission if partially misaligned, abrupt halt if fully misaligned	Ensure circlips are secure before testing
Chain derailment (primarily 1 st chain)	Driver and bystanders	Will abruptly halt transmission, and rotating chain could shoot off and harm nearby bystanders	Trial chain with motor off, stack penny washers underneath motor or remove chain links to raise tension if necessary

Appendix D: Measurement Report (Table 14 and 15):

D.1 Shaft 2

Table 14. Shaft 2 measurement report

Dimension ID	Value and Tolerance	Measuring device	Measured Values and Uncertainty /mm	Comments
∅ Bearing	17 ±0.04	Micrometer	17.01 ±0.01	Within tolerance
∅ Gear (20)	20 ±0.02	Micrometer	20 ±0.01	Within tolerance
∅ Nonfunctional	22 (Tolerance not needed)	Micrometer	22.03 ±0.01	N/A
∅ Sprocket (38)	14 ±0.02	Calliper	13.99±0.02	Within tolerance
Bearing Length	34 ±0.1	Ruler	34.05 ±0.10	Within tolerance
Shaft length	204.75 ±0.4	Ruler	204.62 ±0.10	Within tolerance
Keyway Length	5 ±0.2	Calliper	5.04 ±0.02	Within tolerance
Keyway Depth	3 ±0.1	Calliper	2.96 ±0.02	Within tolerance

D.2 Gear

Table 15. 60T gear measurement report

Dimension ID	Value and Tolerance	Measuring device	Measured Values and Uncertainty /mm	Comments
∅ Hub	60 (not needed)	Calliper	60.02 ±0.02	Within tolerance
∅ Outside	124 (not needed)	Calliper	123.97 ±0.02	Within tolerance
∅ Gear	20 ±0.04	Calliper	20.02 ±0.02	Within tolerance
Face Width	20(not needed)	Calliper	19.98 ±0.02	N/A
Outside Width	10(not needed)	Calliper	10.03 ±0.02	Within tolerance
Keyway Length	5 ±0.2	Calliper	4.94 ±0.02	Within tolerance
Keyway Depth	2.3 ±0.1	Calliper	2.28 ±0.02	Within tolerance